(FP1) 1. Roots of polynomial equations HW

NO. 1

The cubic equation $x^3 + bx^2 + cx + d = 0$, where b, c and d are constants, has roots α, β, γ. It is given that $\alpha\beta\gamma = -1$.

a. State the value of d. [1]

b. Find a cubic equation, with coefficients in terms of b and c, whose roots are $\alpha+1$, $\beta+1$, $\gamma+1$. [3]

c. Given also that $\gamma + 1 = -\alpha - 1$, deduce that $(c - 2b + 3)(b - 3) = b - c$. [4]

NO. 2

The cubic equation $7x^3 + 3x^2 + 5x + 1 = 0$, has roots α, β, γ.

a. Find a cubic equation whose roots are $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$. [3]

b. Find the value of $\alpha^{-2} + \beta^{-2} + \gamma^{-2}$. [2]

c. Find the value of $\alpha^{-3} + \beta^{-3} + \gamma^{-3}$. [2]

NO. 3

The roots of the cubic equation $x^3 + 2x^2 - 3 = 0$ are α, β and γ.

i. By using the substitution $y = \frac{1}{x^2}$, find the cubic equation with roots $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$. [3]

ii. Hence find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. [1]

iii. Find also the value of $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\gamma^2\alpha^2}$. [1]

NO. 4

By finding a cubic equation whose roots are α, β and γ, solve the set of simultaneous equations

$$\begin{aligned} \alpha + \beta + \gamma &= -1, \\ \alpha^2 + \beta^2 + \gamma^2 &= 29, \\ \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= -1. \end{aligned}$$

[8]

NO. 5

i. The cubic equation

$$z^3 - z^2 - z - 5 = 0$$

has roots α, β and γ. Show that the value of $\alpha^3 + \beta^3 + \gamma^3$ is 19. [4]

ii. Find the value of $\alpha^4 + \beta^4 + \gamma^4$. [2]

iii. Show that the cubic equation with roots $\frac{\alpha - 1}{\alpha}$, $\frac{\beta - 1}{\beta}$ and $\frac{\gamma - 1}{\gamma}$ may be found using the substitution $z = \frac{1}{1 - x}$, and find this equation, giving your answer in the form $px^3 + qx^2 + rx + s = 0$, where p, q, r and s are constants to be determined. [4]

NO. 6

The cubic equation $x^3 + bx^2 + cx + d = 0$, where $d \neq 0$, has roots α, β, γ such that $\gamma = \frac{1}{\beta}$.

a. Show that $\beta + \frac{1}{\beta} = d - b$. [3]

b. Show also that $\beta + \frac{1}{\beta} = \frac{1-c}{d}$. [2]

c. It is given that $b = 3$, $c = -3$ and $d > 0$.

i. Find the value of d. [2]

ii. Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

NO. 7

The cubic equation $x^3 + 2x + 1 = 0$ has roots α, β, γ.

a. Find a cubic equation whose roots are $\alpha^3 - 1, \beta^3 - 1, \gamma^3 - 1$. [3]

b. Find the value of $(\alpha^3 - 1)^2, (\beta^3 - 1)^2, (\gamma^3 - 1)^2$. [2]

c. Find the value of $(\alpha^3 - 1)^3, (\beta^3 - 1)^3, (\gamma^3 - 1)^3$. [2]

NO. 8

It is given that

$$\alpha + \beta + \gamma + \delta = 2,$$
$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 3,$$
$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 = 4,$$

a. Find the value of $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$. [2]

b. Find the value of $\alpha^2\beta + \alpha^2\gamma + \alpha^2\delta + \beta^2\alpha + \beta^2\gamma + \beta^2\delta + \gamma^2\alpha + \gamma^2\beta + \gamma^2\delta + \delta^2\alpha + \delta^2\beta + \delta^2\gamma$. [3]

c. It is given that α, β, γ, δ are the roots of the equation

$$6x^4 - 12x^3 + 3x^2 + 2x + 6 = 0.$$

i. Find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$. [3]

ii. Find the value of $\alpha^5 + \beta^5 + \gamma^5 + \delta^5$. [2]

NO. 9

The cubic equation $27x^3 + 18x^2 + 6x - 1 = 0$ has roots α, β, γ.

a. Show that a cubic equation with roots $3\alpha+1, 3\beta+1, 3\gamma+1$ is $y^3 - y^2 + y - 2 = 0$. [3]

The sum $(3\alpha+1)^n + (3\beta+1)^n + (3\gamma+1)^n$ is denoted by S_n.

b. Find the values of S_2 and S_3. [4]

c. Find the values of S_{-1} and S_{-2}. [3]

NO. 10

A cubic equation $x^3 + bx^2 + cx + d = 0$ has real roots α, β and γ such that

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{5}{12},$$

$$\alpha\beta\gamma = -12,$$

$$\alpha^3 + \beta^3 + \gamma^3 = 90.$$

i. Find the values of c and d. [3]

ii. Express $\alpha^2 + \beta^2 + \gamma^2$ in terms of b. [2]

iii. Show that $b^3 - 15b + 126 = 0$. [4]

iv. Given that $3 + i\sqrt{(12)}$ is a root of $y^3 - 15y + 126 = 0$, deduce the value of b. [2]